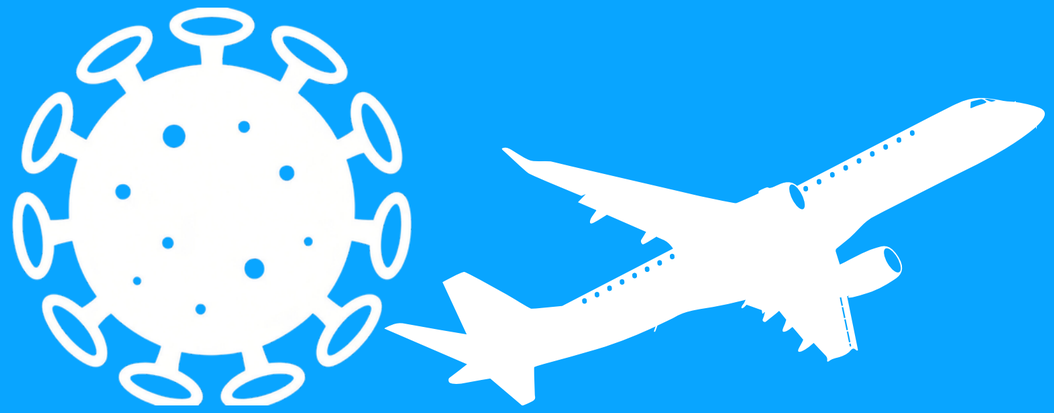


ANALYSIS OF AN AIRPORT NETWORK DISEASE SPREAD MODEL

Marco Minchella

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CONTENT ROADMAP

Research questions

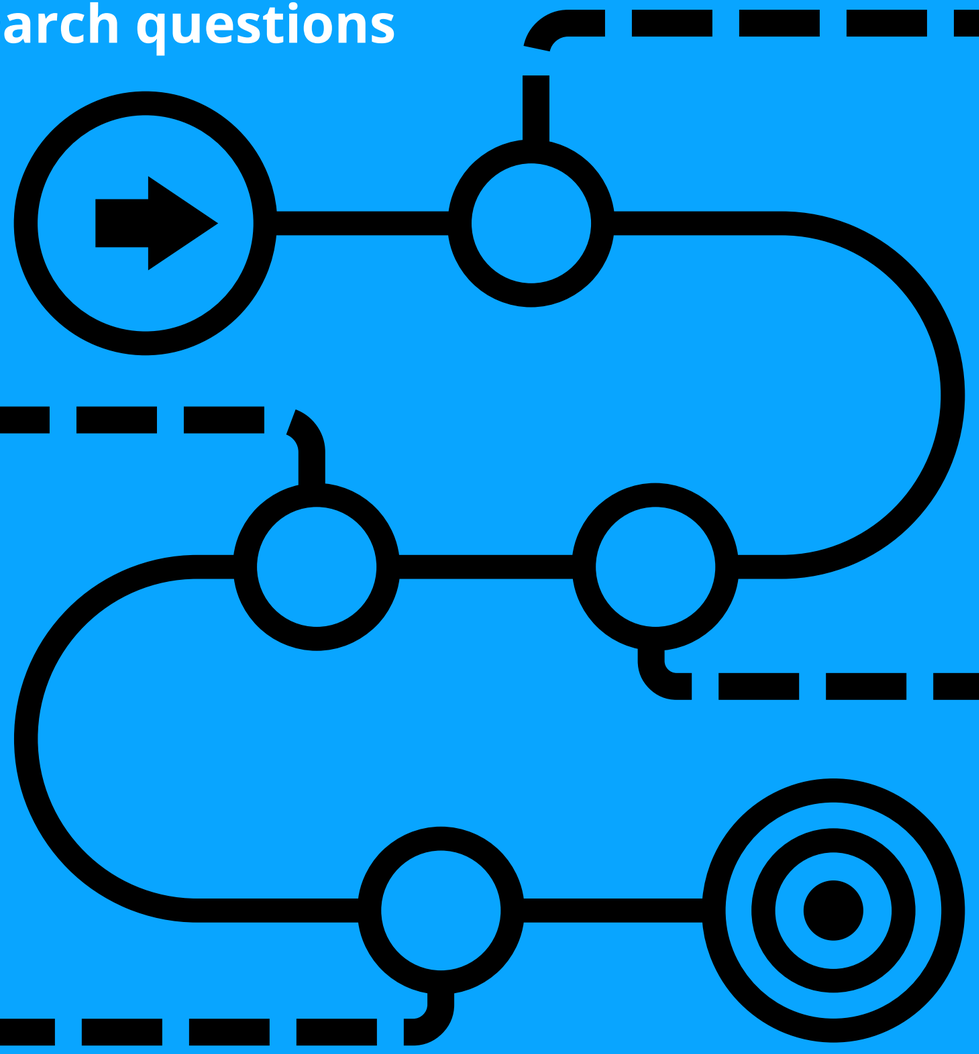
The network:
dataset, degree distributions, centralities, assortativity, robustness

The epidemic model:
SEIR model on the WAN

Community structure:
Louvain and InfoMap

Vaccination strategies :
Strategies results analyisis

Conclusions

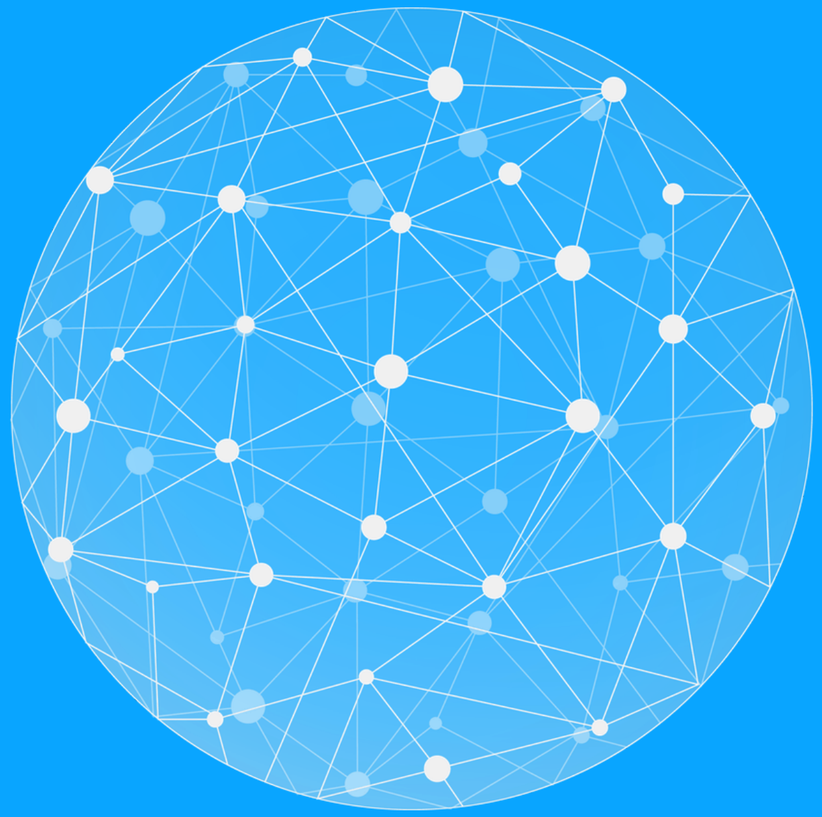


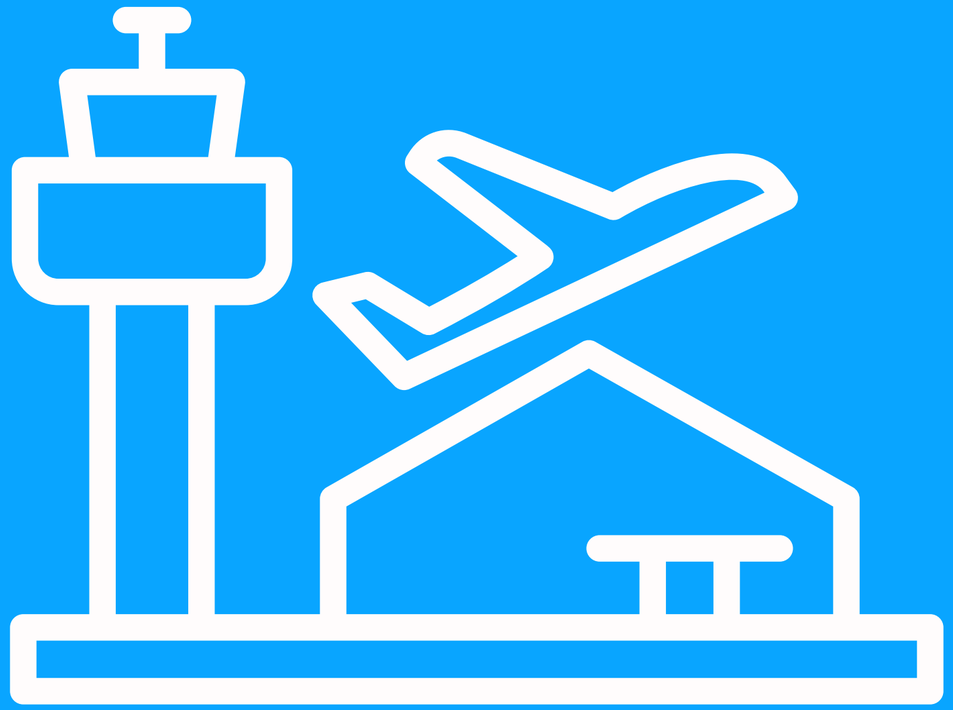
WHAT ARE THE RESEARCH QUESTIONS?



- What are the **characteristics of the World Airport Network?** Does it have any **interesting properties?**
- Is **hub targeting** the best way to obstruct a disease spread? Are there **other preferable approaches?**
- How do our results change if we factor in the **economic aspect** of shutting down entire airports?

THE NETWORK





WHY THE GLOBAL AIRPORT NETWORK?

- Air transport is defined as a major channel for the spread of epidemics [1]

- Airport networks are also used as benchmarks to verify whether a centrality measure really works [2]

- On a global scale, the WAN is the primary driver of viral seeding across distant urban hubs. Given its massive throughput and point-to-point topology, air traffic data provides a high-fidelity approximation for long-range epidemic forecasting [3]



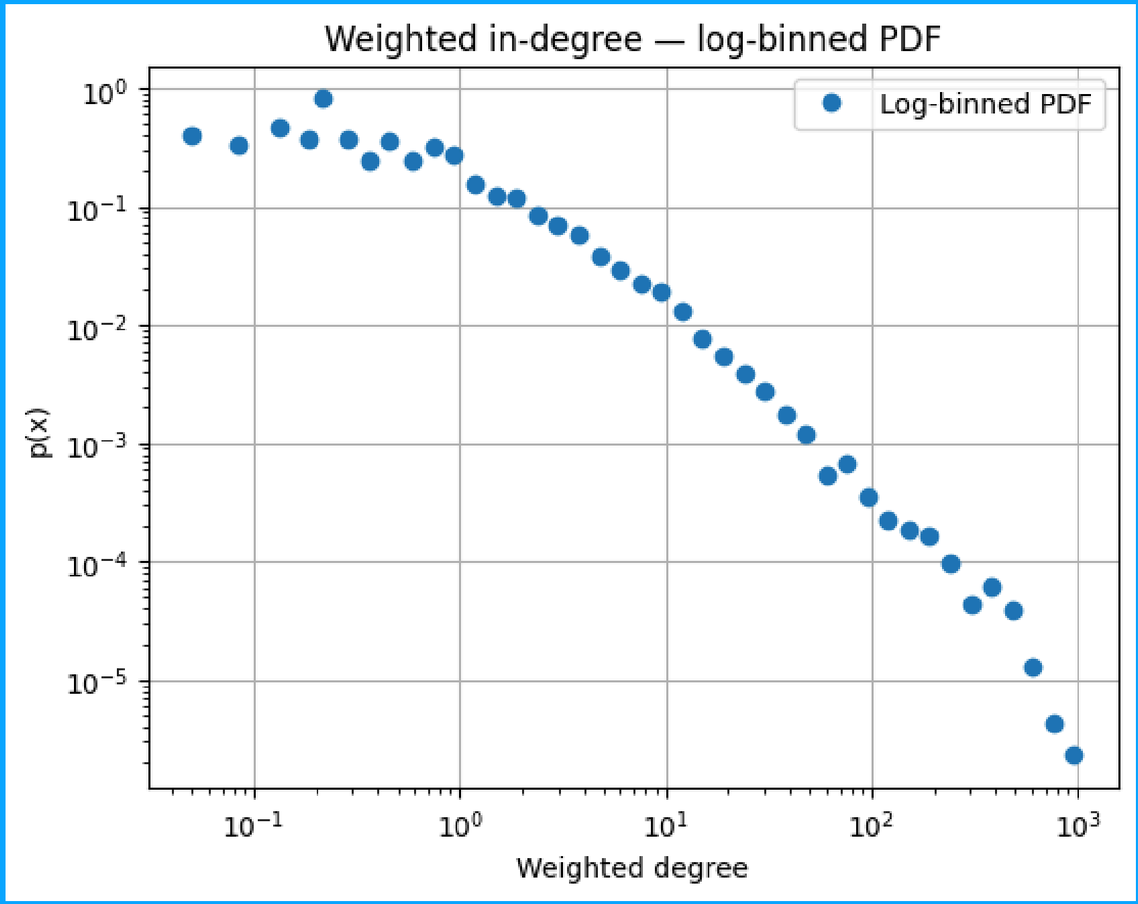
THE DATASET

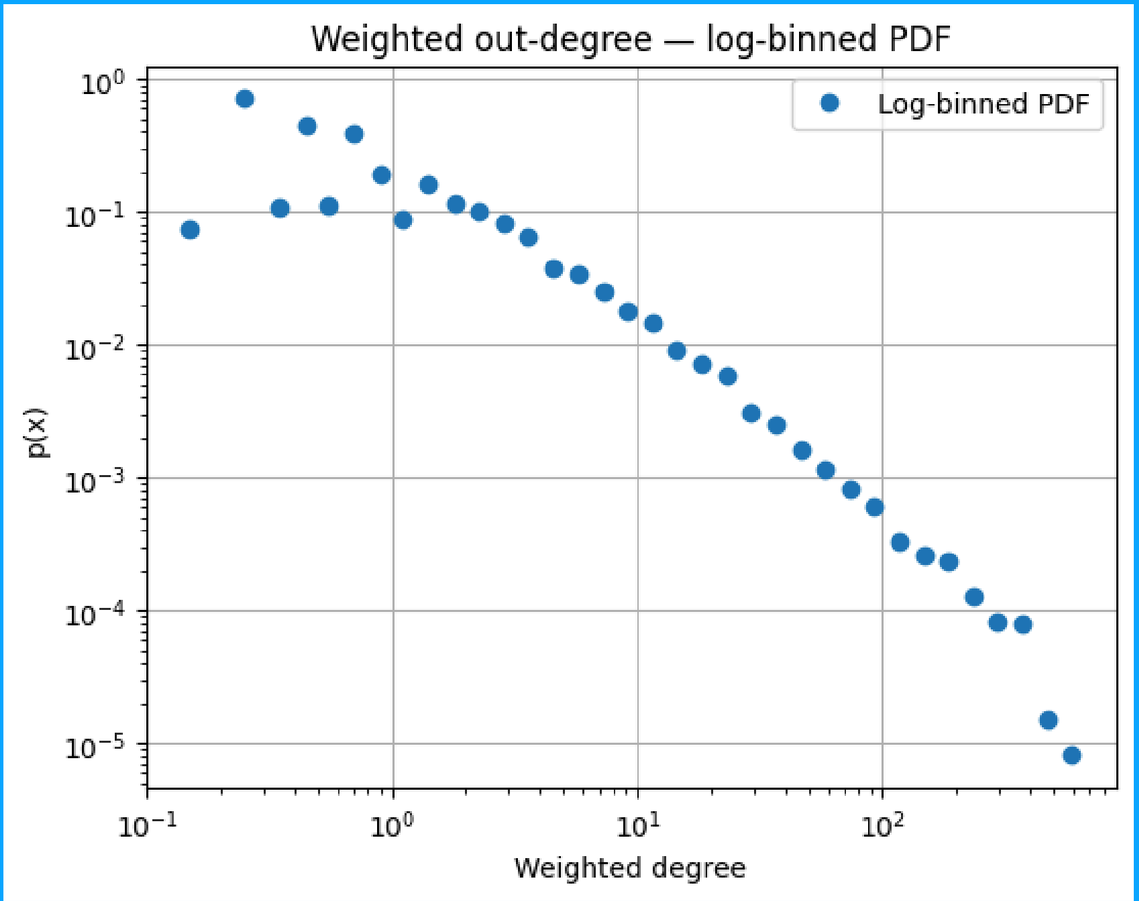


- Airports & routes taken from OpenFlights.org
- Mean daily flights taken from FlightConnections.com
- Airports/city population taken from WorldPopulationReview.com

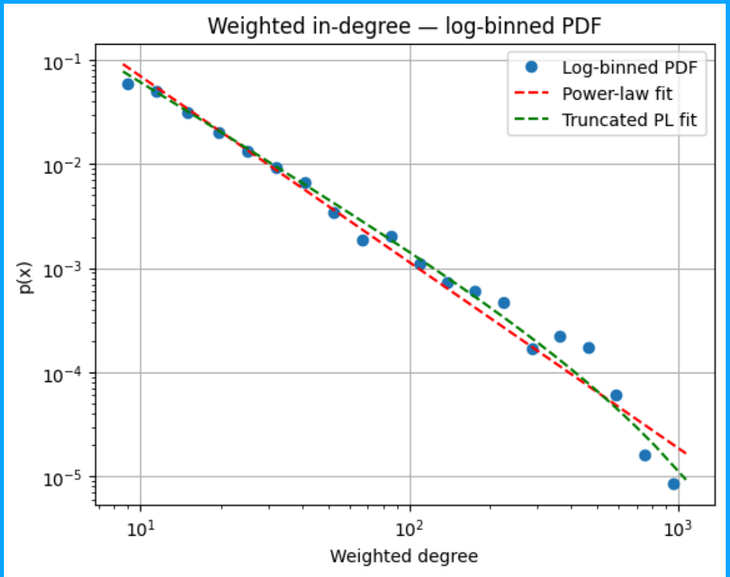
NODES	EDGES	POPULATION
3933	50158	2.685.490.062

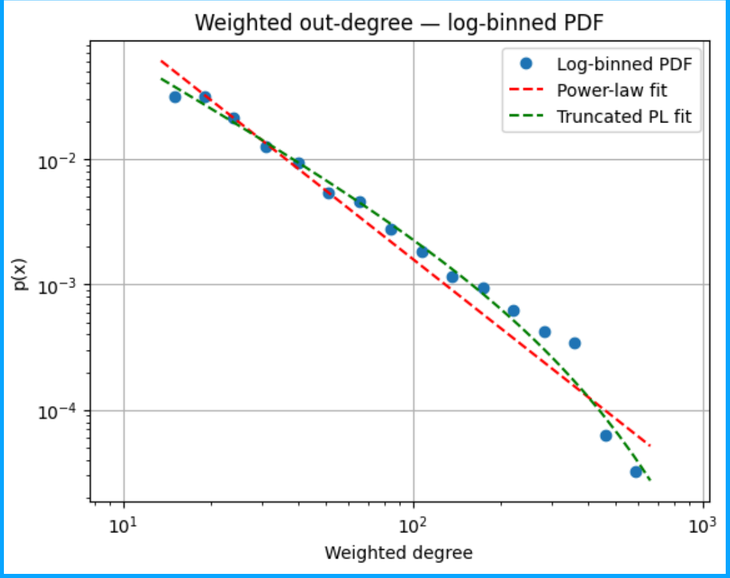
DEGREE CENTRALITIES DISTRIBUTIONS





DEGREE CENTRALITIES DISTRIBUTIONS





What do these graphs tell us?

Both weighted degree distributions are heavy-tailed, but neither follows a pure power law

$$p(x) = Cx^{-\gamma}$$

Likelihood-ratio tests favour a **truncated power law** in both directions, more decisively for out-strength (R = −48.4) than for in-strength (R = −23.7)

$$p(x) = Cx^{-\gamma}e^{-\lambda x}$$

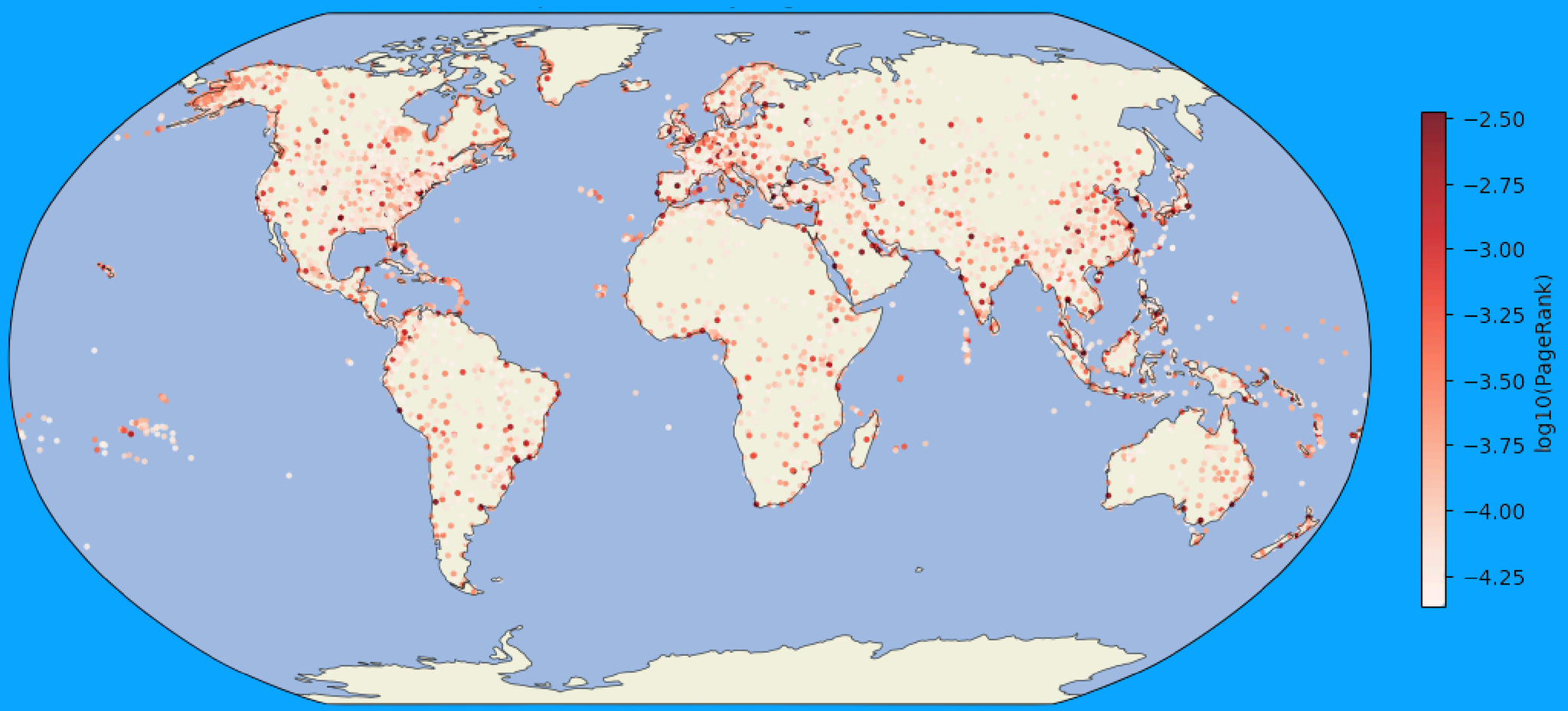
	x min	gamma (pure PL)	gamma (truncated PL)	k_c (truncated PL)
Weighted in-degree	**8,6**	**1,787**	**1,58**	**1/0,0013 = 750**
Weighted out-degree	**13,5**	**1,817**	**1,328**	**1/0,0034 = 292**

Why does this happen?

Heavy-tailed degree distributions are expected (i.e. **demand concentration in major metropolitan areas**), but there are **strong real-world constraints** that introduce a cutoff:

- **Finite capacity** at airports (runway slots, gates, crew and fleet limitations, etc)
- **Scheduling** and **regulatory constraints**
- Demand and operational **efficiency saturate**

PAGERANK



The majority of high PageRank airports tend to be clustered in North America, Europe and East Asia

PAGERANK

Rank	Airport	PageRank (10^-3)
1	Chicago O'Hare International Airport	7.125
2	Hartsfield Jackson Atlanta International Airport	6.912
3	Istanbul Airport	6.323
4	Los Angeles International Airport	5.639
5	Dallas Fort Worth International Airport	5.595
6	Amsterdam Airport Schiphol	5.573
7	Charles de Gaulle International Airport	5.459
8	Denver International Airport	5.427
9	Sydney Kingsford Smith International Airport	5.372
10	London Heathrow Airport	5.339
11	Sheremetyevo International Airport	5.152
12	El Dorado International Airport	5.090
13	Frankfurt am Main Airport	4.887
14	Indira Gandhi International Airport	4.855
15	Adolfo Suárez Madrid–Barajas Airport	4.767

PageRank can be interpreted as:
“a node is important if it is linked to by important nodes”

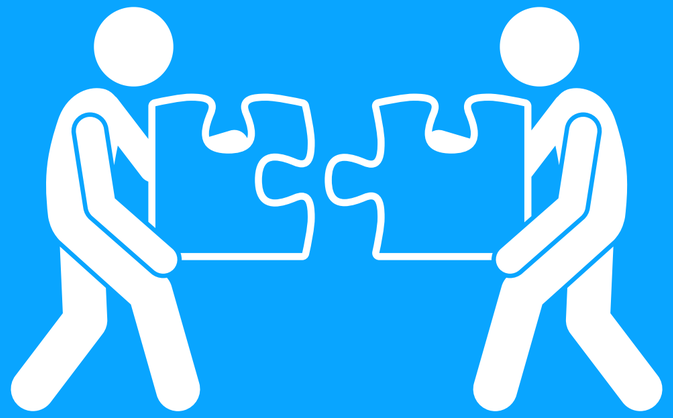
Reasonable proxy for arrival risk and for identifying “waypoints” in global spread

Is it the best measure to estimate which nodes should be removed?
It’s a good measure, but literature shows we can do better...[4]

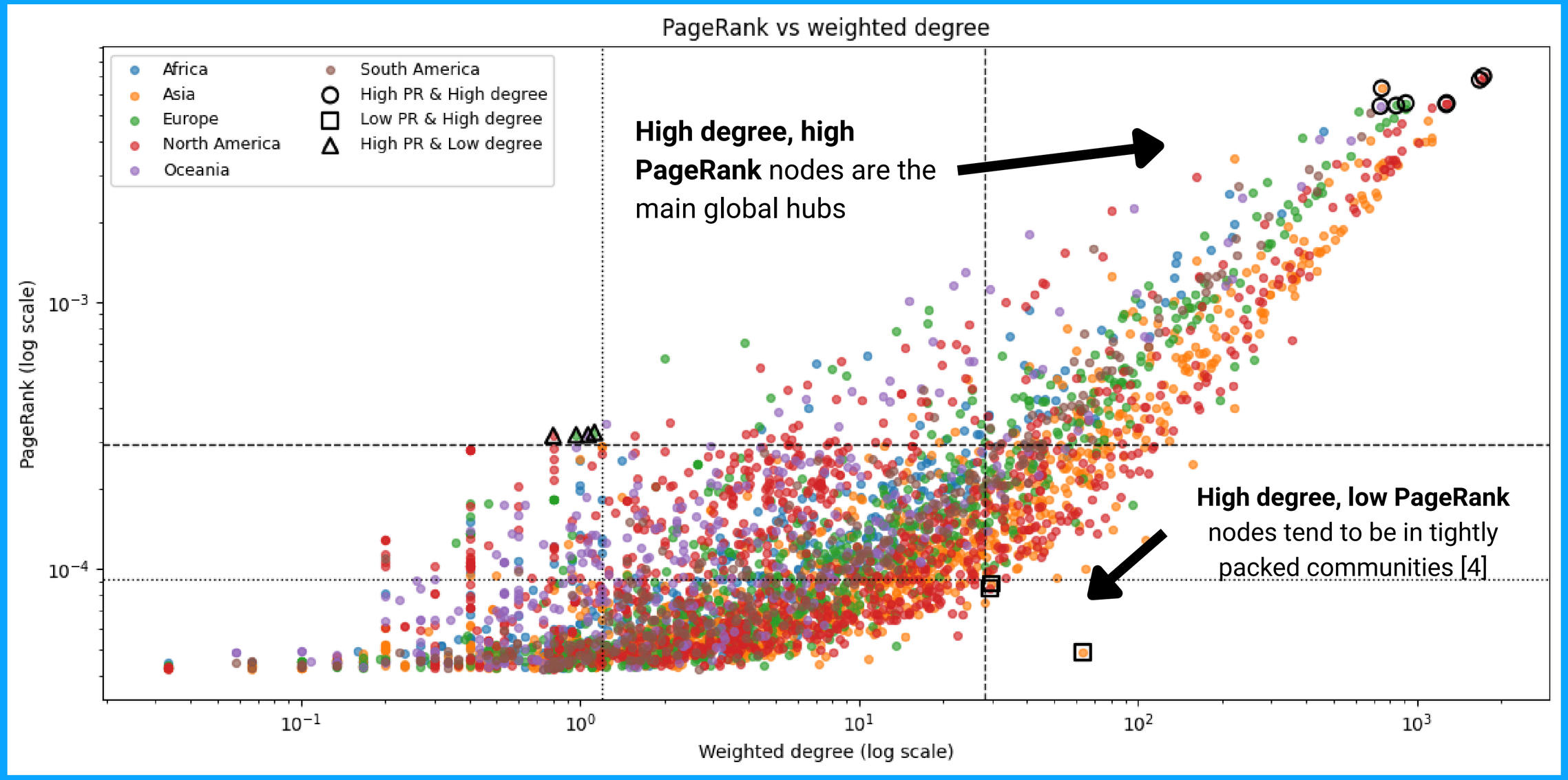
PAGERANK AND WEIGHTED DEGREE

Does it make sense to combine these two measures?

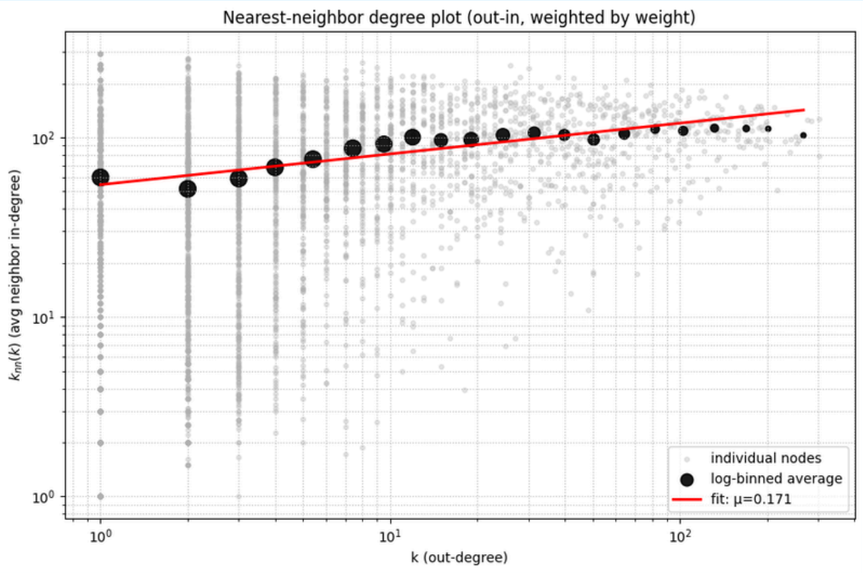
- Weighted degree is a purely local measure. PageRank propagates importance through the whole network.
- **Idea**: take $s_i = \tilde{p}_i \cdot \tilde{w}_i$, the **product of min-max normalised weighted PageRank and weighted degree**. Look at airports that are both high-volume and structurally central
- Literature shows it's **one of the most powerful combinations** for predicting the absolute size of the outbreak[4]

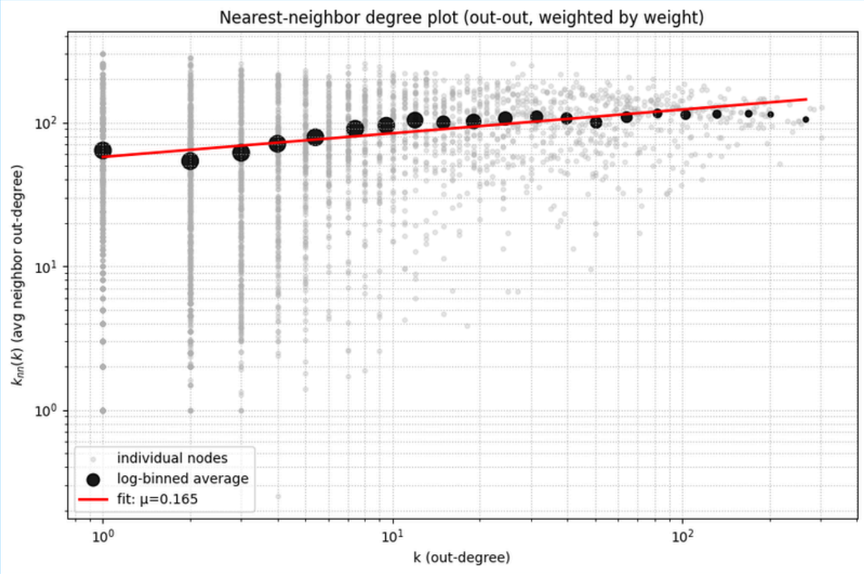


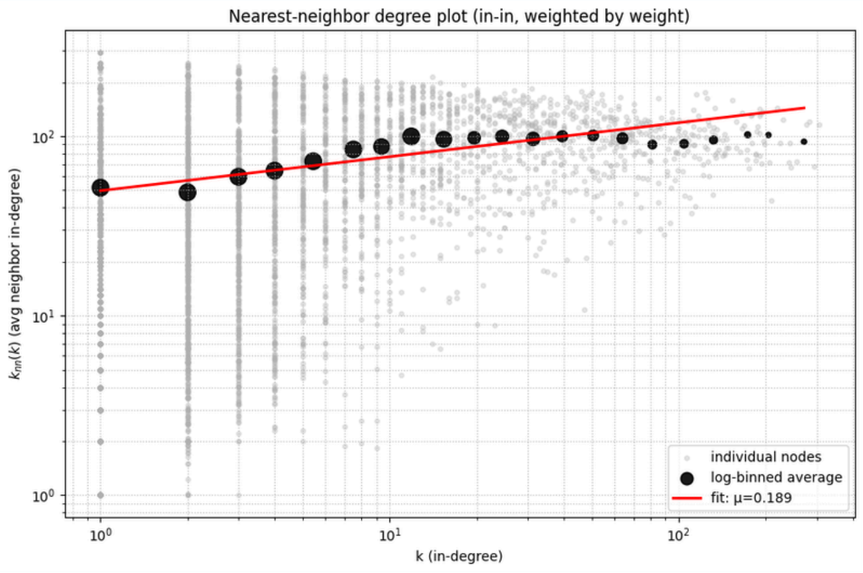
PAGERANK AND WEIGHTED DEGREE

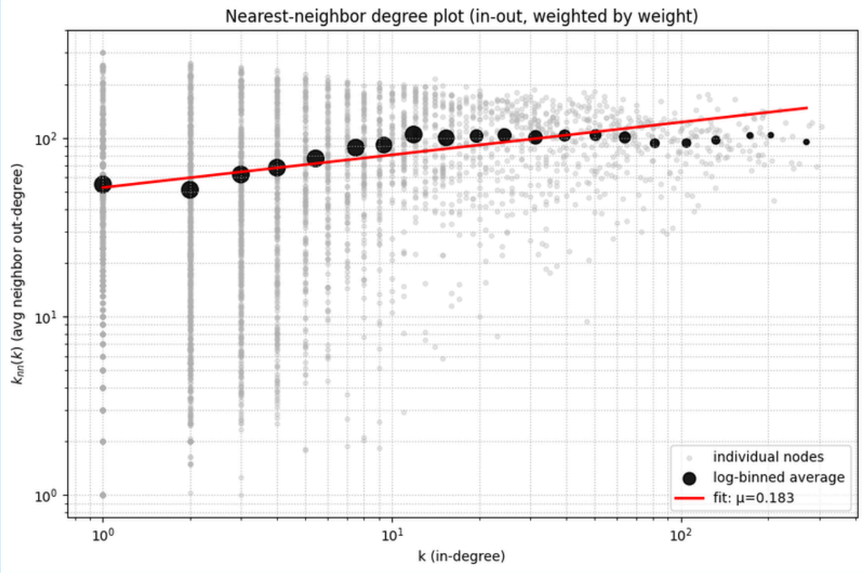


ASSORTATIVITY









ASSORTATIVITY – A CLOSER LOOK

What do the nn degree plots tell us?

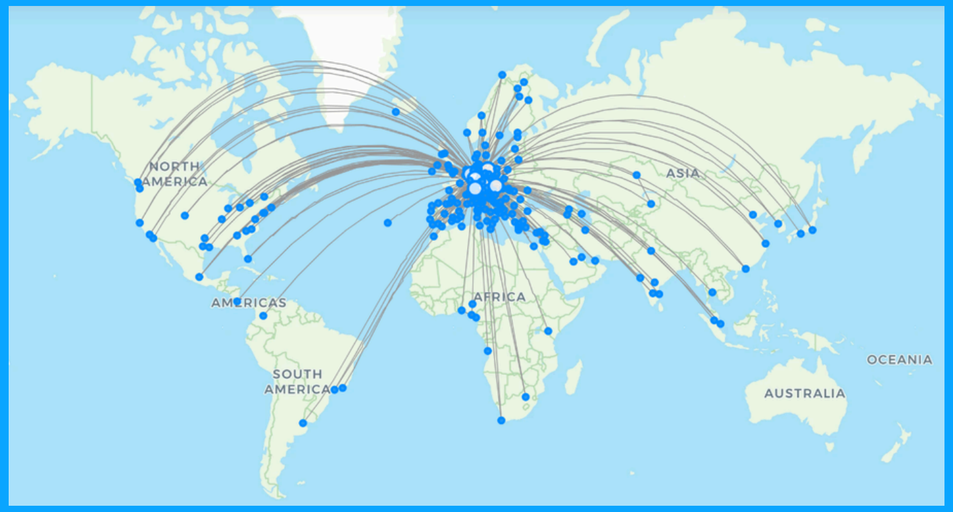
- All fitted exponents are positive, with **μ⩾0.165**
- The network is **assortative**

But is the whole network assortative?

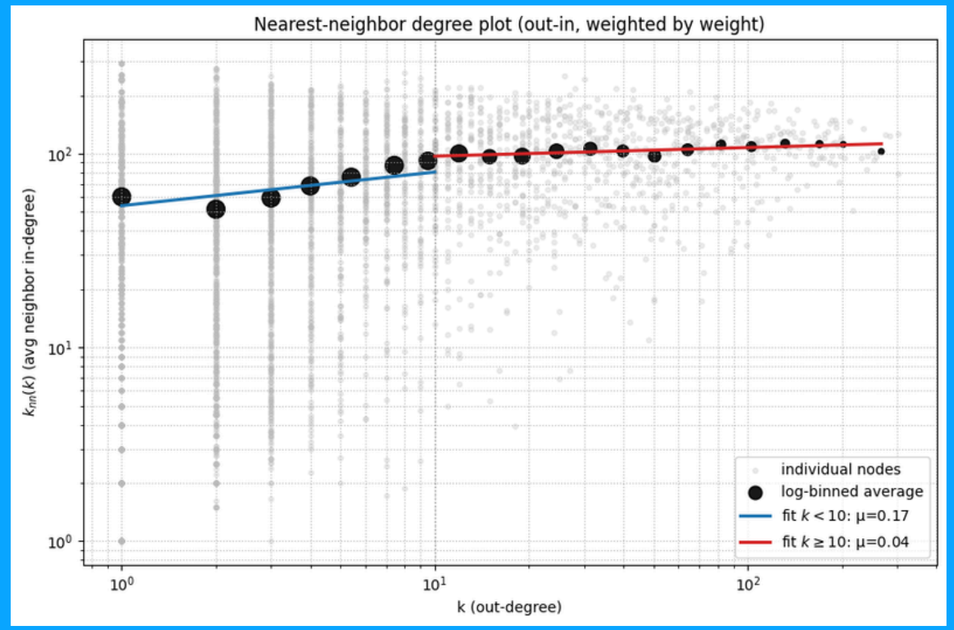
- Restricting the fit to k≥10 (major hubs) gives μ≈0.04 i.e. essentially flat
- The positive slope is driven almost entirely by airports with k<10

Why?

- Small/medium airports connect preferentially to similarly-sized peers → the network behaves like an assortative mesh of highly interconnected regional and domestic communities
- Major hubs must connect to airports of highly variable size to fulfill their collecting role → this produces a plateau
- Consistent with *Barrat et al. (The architecture of complex weighted networks, 2004)*

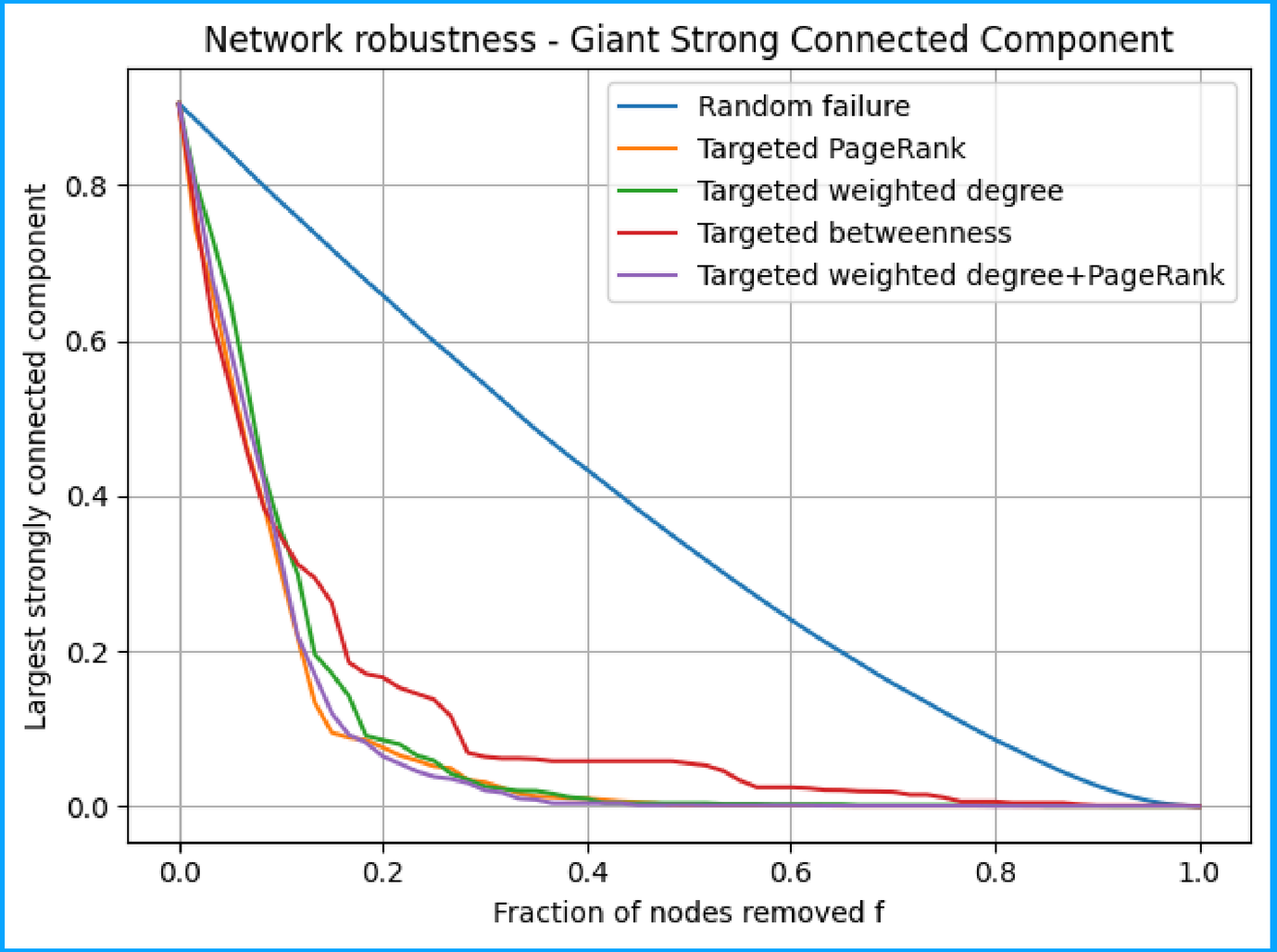


Lufthansa's air routes: they connect to other hubs (i.e. LAX) as well as small airports



LET’S BREAK THE NETWORK: ROBUSTNESS ANALYSIS

- GSCC starts at 0.905, around 370 airports lie outside the giant strongly connected component
- Targeted attacks: f ≈ 0.15 collapses the core, except betweenness, which leaves a residual periphery up to f ≈ 0.65
- The residual core under betweenness attack is the **assortative low-degree mesh** seen in assortativity analisys

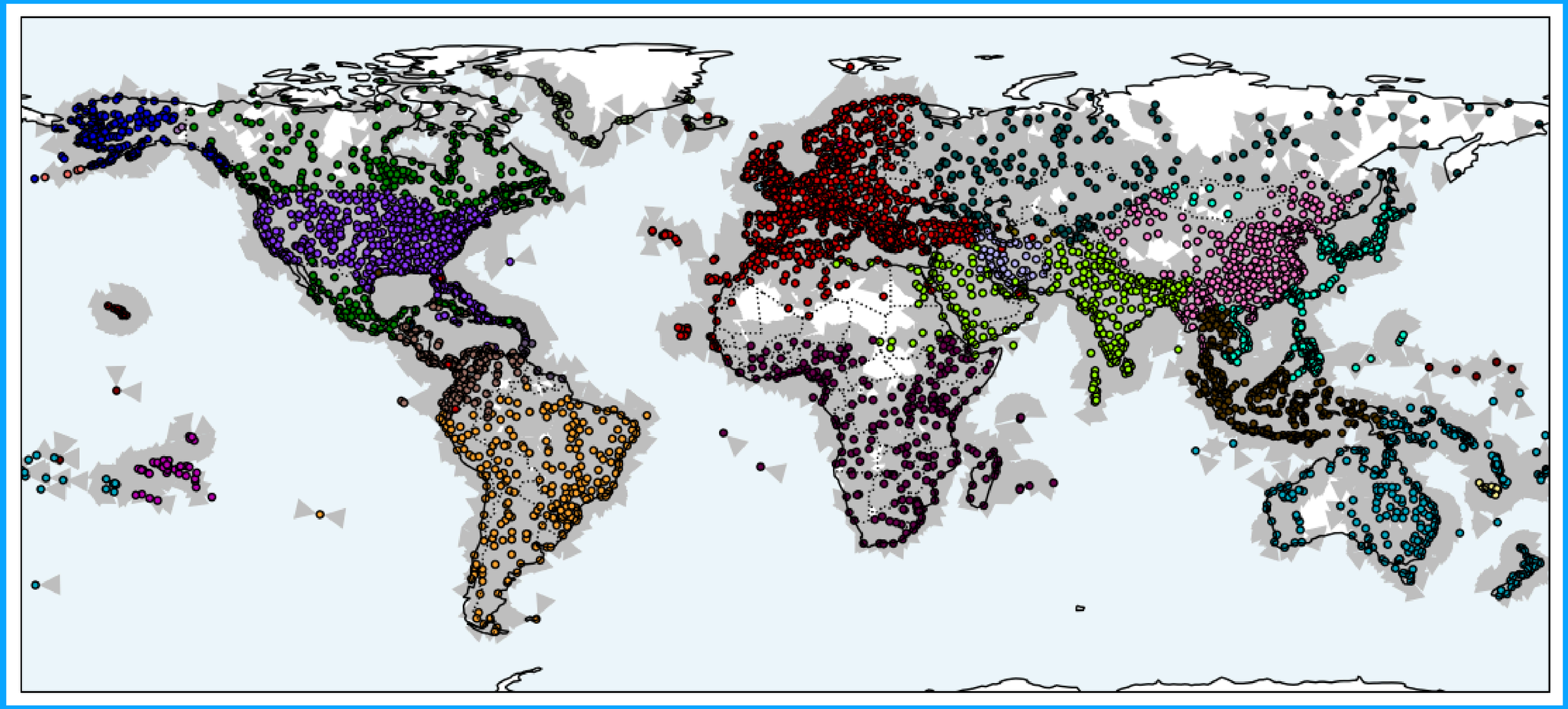


COMMUNITY STRUCTURE



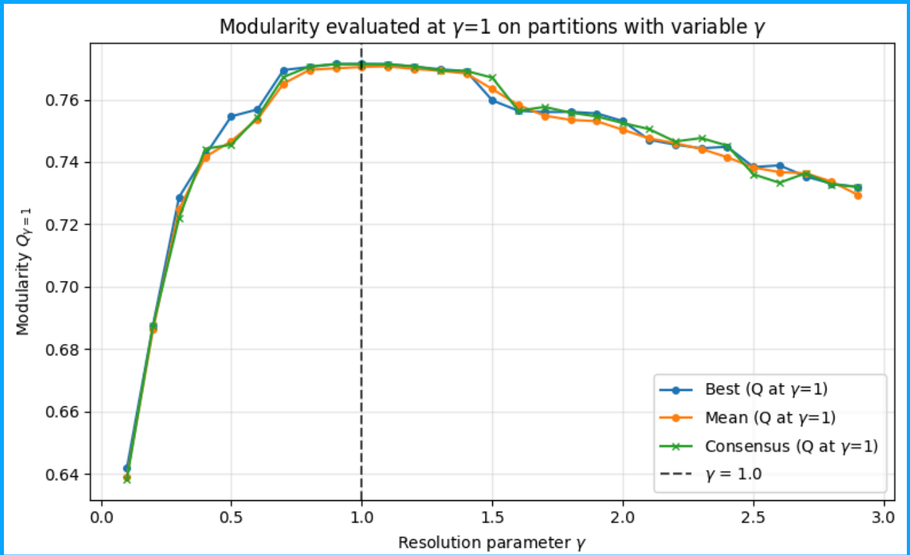


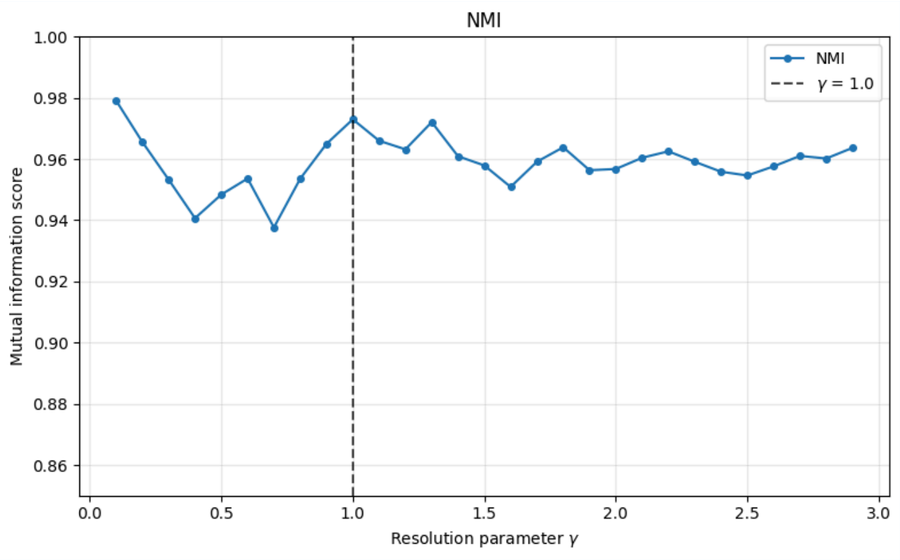
COMMUNITY DETECTION: LOUVAIN



Communities align with geographic and geopolitical regions

COMMUNITY DETECTION: LOUVAIN

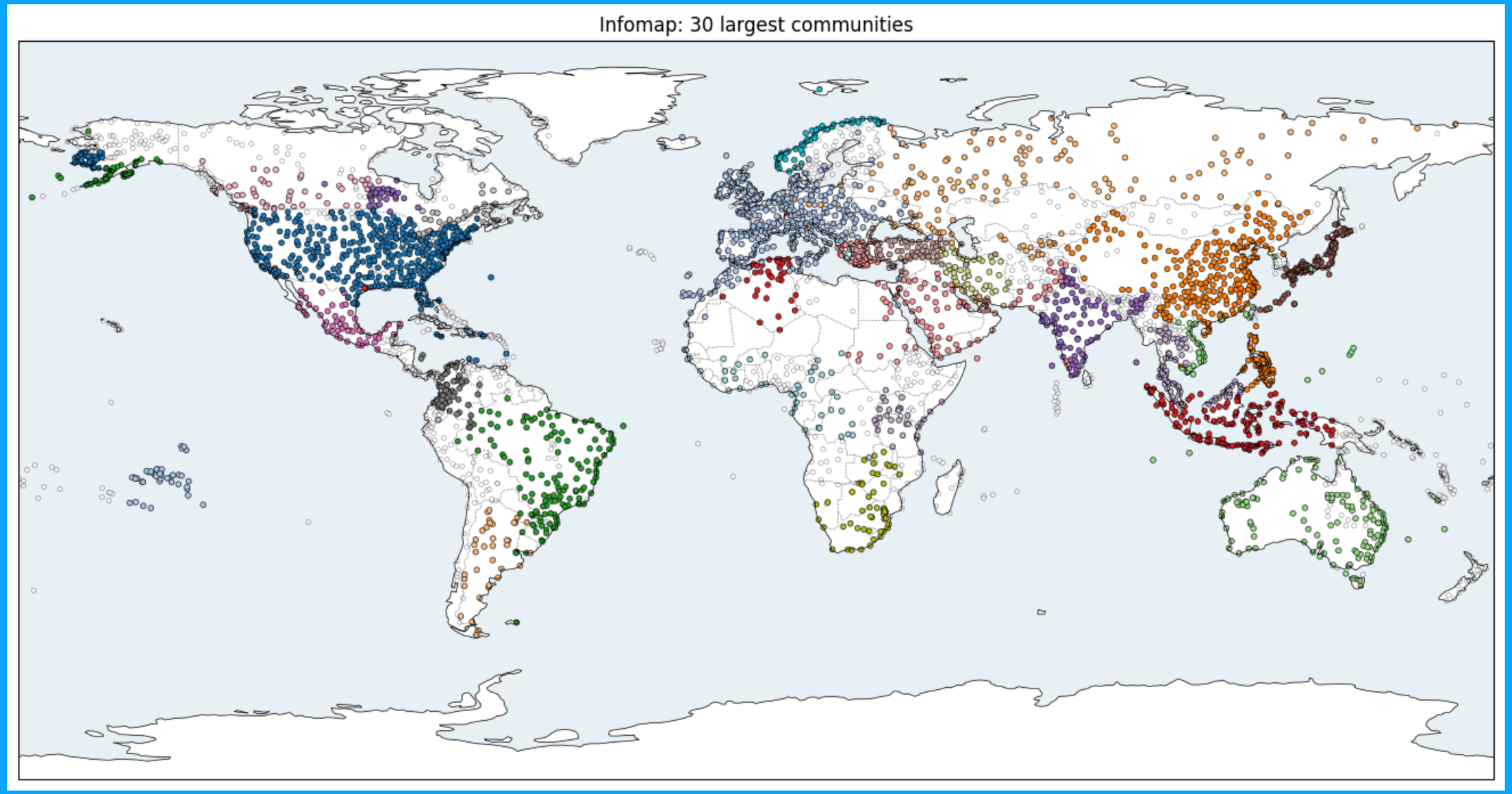




To select the optimal resolution parameter, **consensus clustering** was employed with 50 iterations. Observing the modularity score we get its peak value at a resolution value of 1, while also exhibiting a local peak in NMI.

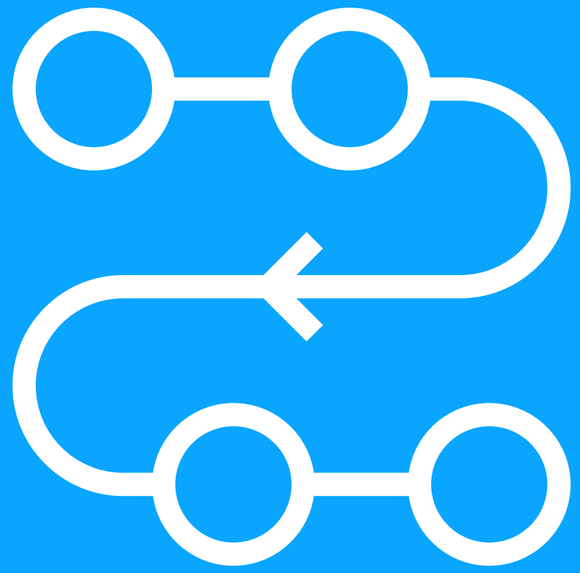
COMMUNITIES	30
MODULARITY	0.7714
CODELENGTH	6.9195
NMI (stability)	0.9767

COMMUNITY DETECTION: INFOMAP



COMMUNITY DETECTION: INFOMAP

Minimises the map equation: cost of describing a random walk with a two-level code.
Modules are regions where flow gets trapped. They are **plausible epidemic basins** .
Consensus clustering was employed with the same number of runs as in Louvain.



COMMUNITIES	148
MODULARITY	0.7312
CODELENGTH	6.5250
NMI (stability)	0.9859

LOUVAIN VS INFOMAP



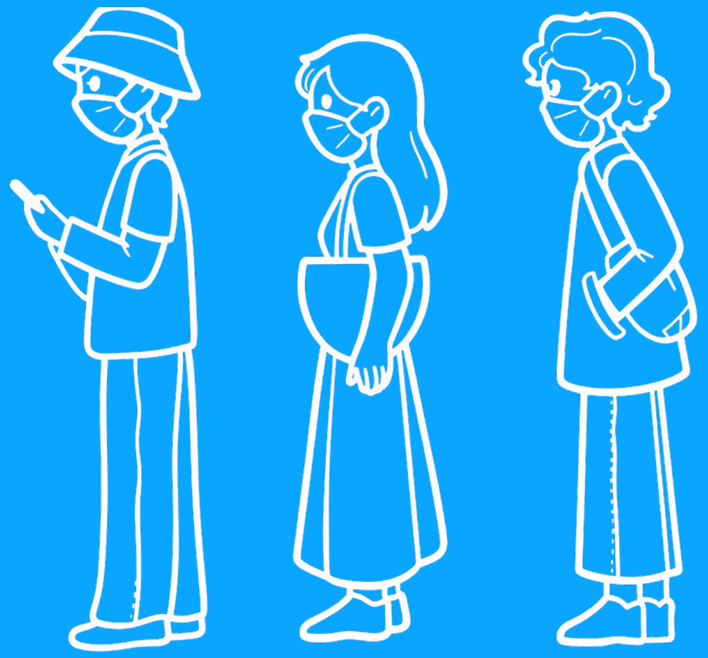
	Louvain	InfoMap
COMMUNITIES	30	**148**
MODULARITY	**0.7714**	0.7312
CODELENGTH	6.9195	**6.5250**
NMI (stability)	0.9767	**0.9859**

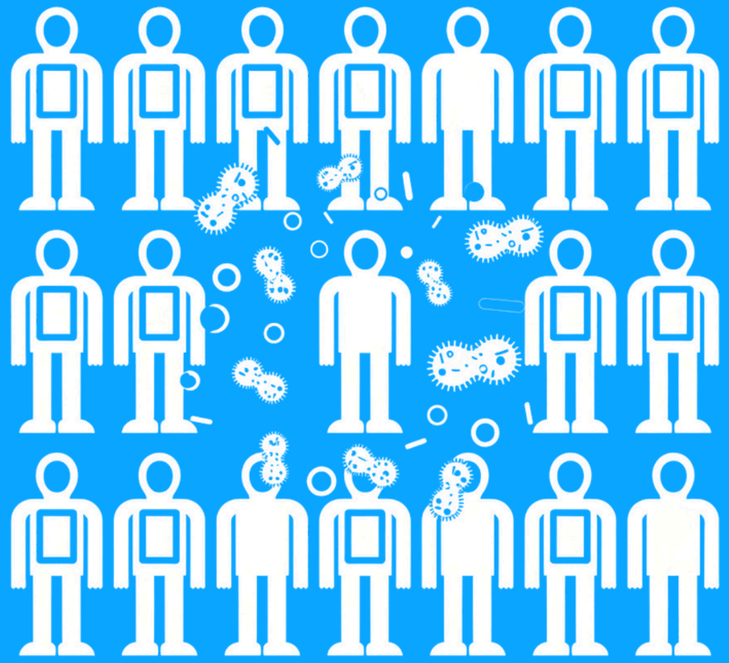


As expected **Louvain wins on modularity** while InfoMap **minimizes the codelength.**

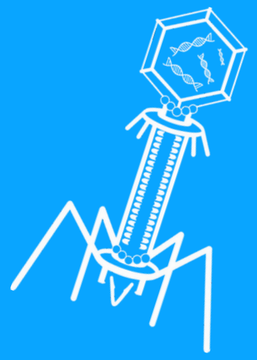
These communities results will be used to drive vaccination strategies...

THE EPIDEMIC MODEL

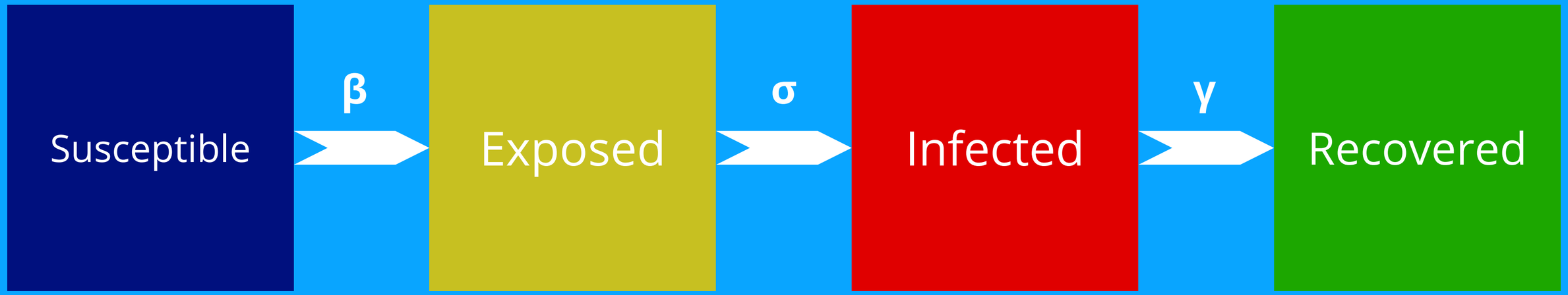




THE EPIDEMIC MODEL



SEIR model



R0 = Basic reproduction number. Key value, higher rate means increased aggressiveness

β = Transmission rate ≈ γ*R0

σ = Infection rate = 1/Latent period

γ = Recovery rate = 1/Infectious period

THE EPIDEMIC MODEL

Stochastic SEIR model: breakdown on a single node

- **The Probability**: Continuous rates (like β or γ) are converted into discrete probabilities using

$$1 - e^{-\beta \frac{I}{N} dT}$$

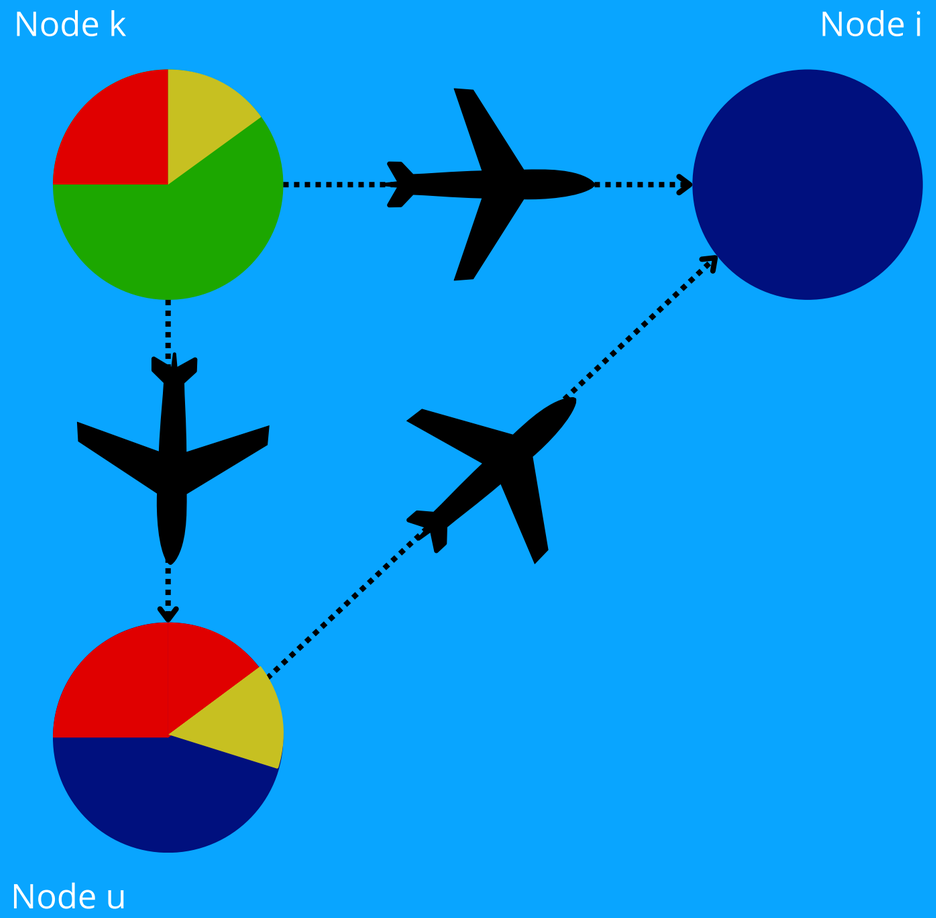
This calculates the chance of a single individual transitioning within the time step dT.

- **The Usage**: These probabilities act as the "success rate" in a Binomial distribution. The model treats each subsceptible as an independent trial.

- **Why Integers**: Because the Binomial distribution samples a specific count of successes, the output is always a whole number. This prevents "fractional people" and captures the natural randomness (stochasticity) of real-world outbreaks.

THE EPIDEMIC MODEL

Stochastic SEIR model: inter-nodes infection



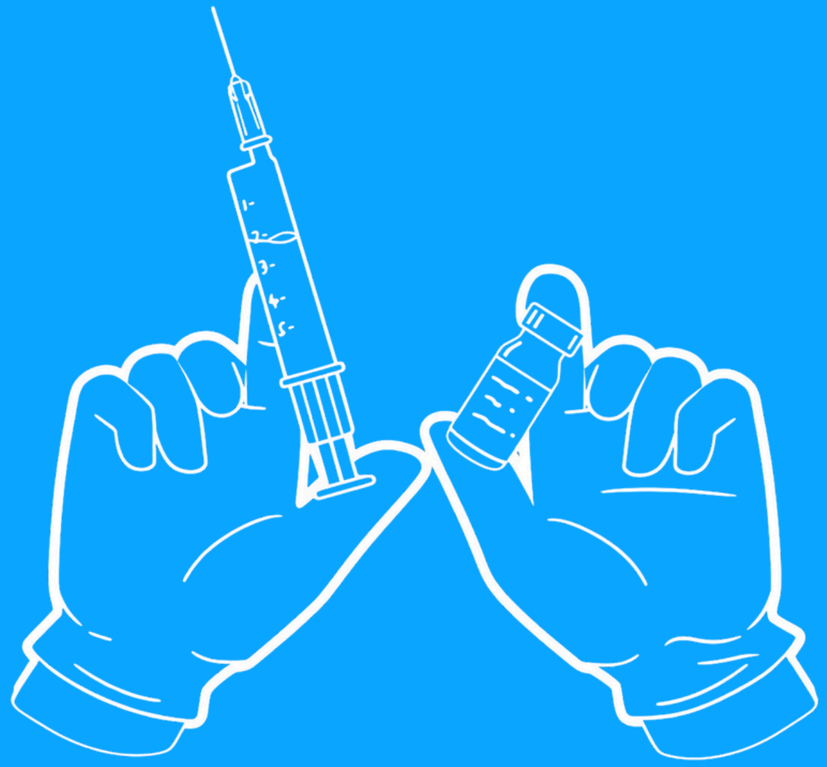
- Each infected node (I>0) has a certain probability of infecting successor nodes with at least one susceptible individual
- Daily flights are treated in 2 different ways
  1. Taken from Poisson distribution with mean equal to the edge's weight if mean flights are less than 20
  2. Treated as a deterministic value but with a probability of cancellation of 2% if mean flights are equal or more than 20
- *P(Node i gets infected)* is calculated as follows:

$$P(\text{Infected Flight}) = \frac{I}{N}$$

$$P(\text{No Infected Flight In A Day}) = (1 - \frac{I}{N})^{\text{Daily Flights}}$$

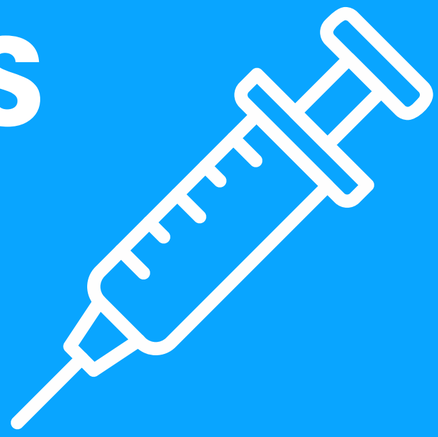
$$P(\text{Node i gets infected}) = 1 - \prod_{j=1}^{\text{infected predececcessors}} P(\text{No Infected Flight In A Day})$$

VACCINATION STRATEGIES





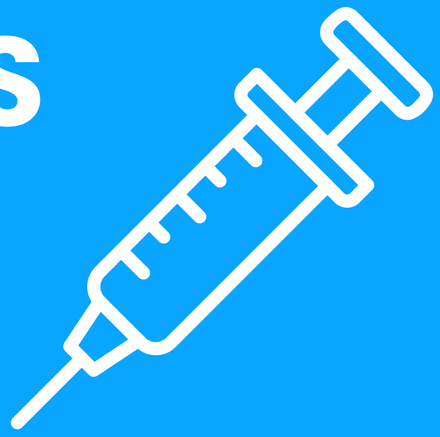
VACCINATION STRATEGIES



Basic hypothesis and rules

1. Vaccinating a node = **Closing down the airport**
2. Each strategy selects **5% of the nodes**
3. Perfect contact tracing
4. Vaccination is **immediately in effect**
5. Flight grounding is the only restriction strategy
6. Economic impact of a strategy is directly proportional to the foreclosed flights. The **cost is calculated on daily grounded planes**.

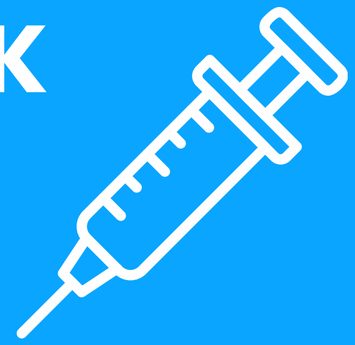
VACCINATION STRATEGIES



The adopted strategies were:

1. **No vaccination:** Serves as baseline
2. **Random:** Nodes were selected at random
3. **Hub:** Targeting nodes with highest weighted degree
4. **Hub + PageRank:** Targeting nodes with highest PR and weighted degree
5. **Betweenness:** Targeting nodes according to their betweenness centrality
6. **Louvain bridges:** Targeting nodes in each community that had many connections to other different communities
7. **Infomap bridges:** same criterion, applied to the Infomap partition
8. **Collective Influence:** Removal is based on scores that measure how much connectivity lies on a node's 2-hop frontier and removes nodes greedily recomputing after each removal.

VACCINATION STRATEGIES – A CLOSER LOOK



How are communities used in this scenario?

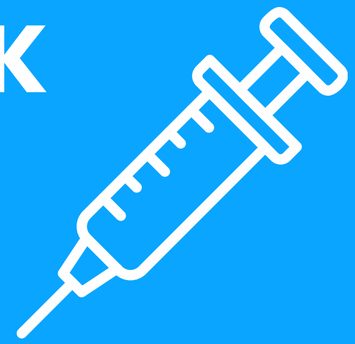
For each airport in a community, compute the Shannon entropy of its outgoing traffic across communities

$$H_i = -\sum_{c}^{C} p_{i,c} \log_2 p_{i,c}$$

where p_i,c is the fraction of out-traffic directed to community c.

Vaccination targets = entropy × total out-traffic, so a bridge only ranks high if it carries meaningful volume and if traffic spread across many communities

VACCINATION STRATEGIES – A CLOSER LOOK



How about the Collective Influence?

It is inspired by the work of *Morone & Makse*, ***Influence maximization in complex networks through optimal percolation (2015)*** adapted to a directed weighted network with the frontier follows outgoing edges only

Scores each airport by the product of its own out-strength and the total out-strength of all nodes exactly 2 hops away

$$CI\left(i\right) = g\left(s_i^{out}\right) \sum_{j \in \partial Ball_{out}(i,2)} g\left(s_j^{out}\right)$$

where s is the weighted out strength and g is the max between s and 0. The non-negativity function is retained from the original formulation, where the original argument (i.e. reduced degree k−1) could vanish

VACCINATION GOALS

1. **Protect** as many nodes as possible
2. **Delay and reduce** the peak
3. **Protect** as much population as possible
4. Select the most **cost efficent** strategy





SIMULATIONS

8 different starting seeds were selected: Chicago, Atlanta, Los Angeles, Istanbul, London, New York JFK, Pago Pago and Uzhorod

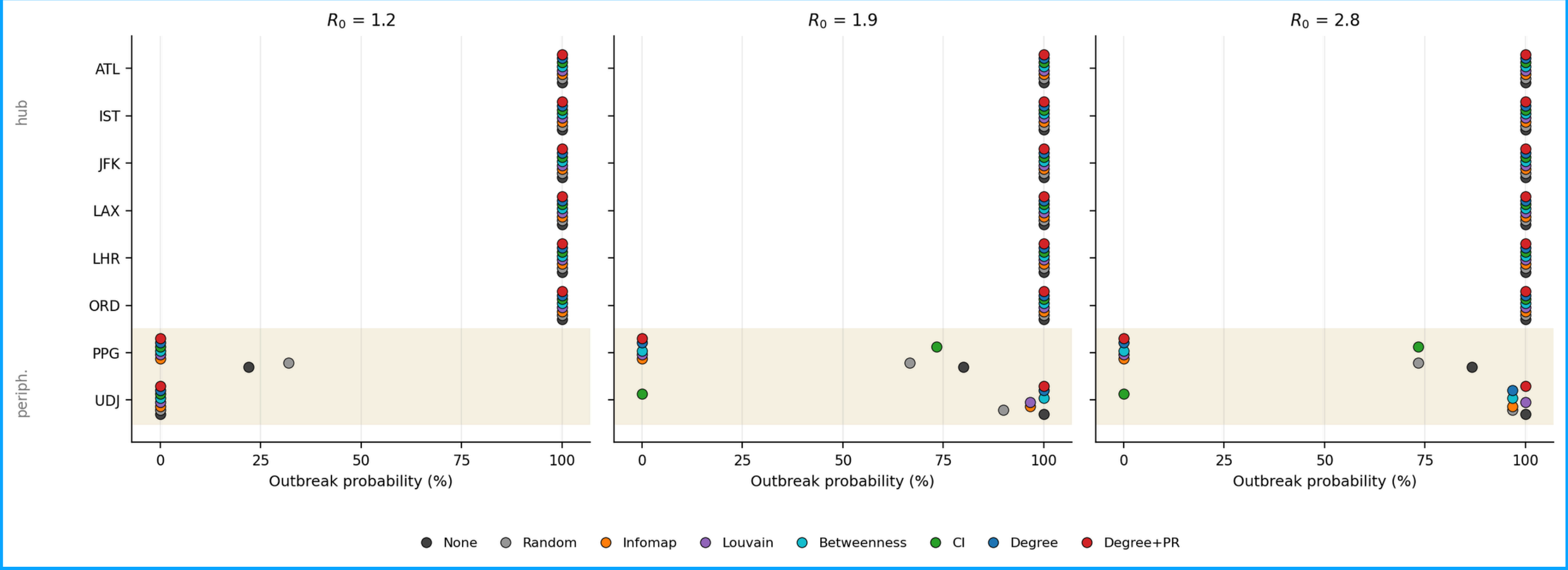
3 different R0 were selected: **1.2, 1.9 and 2,8**

Each combination of seed and R0 was run **30 times**, except for R0 = 1.2, which was run **50 times**

Due to computational limitations, the runs were capped at **548 days** or **1 and a half years**

RESULTS – OUTBREAK PROBABILITY

Outbreak = more than 1% of the nodes is infected

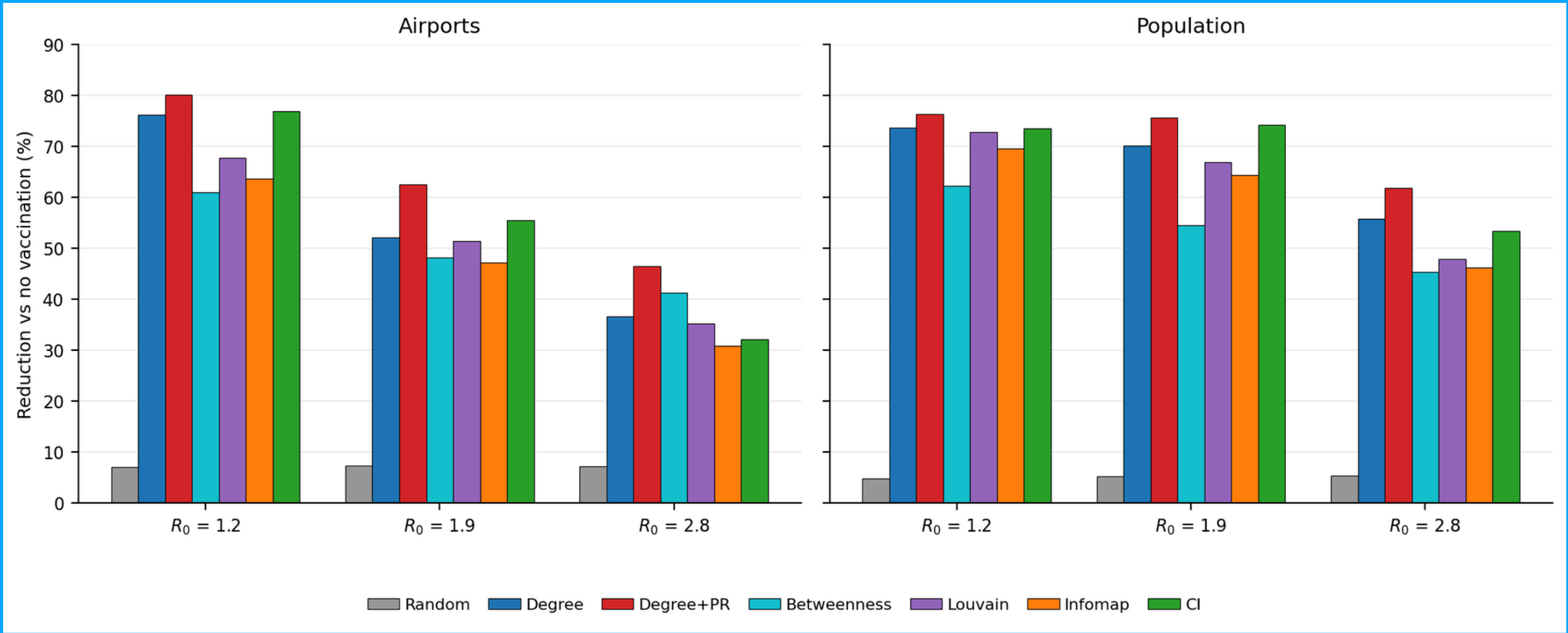


Hub seeds: outbreak certain under every strategy, at every R_0

Peripheral seeds: PPG (worst degree) is suppressed by most strategies while UDJ (worst PR) is largely unaffected, CI does the reverse.

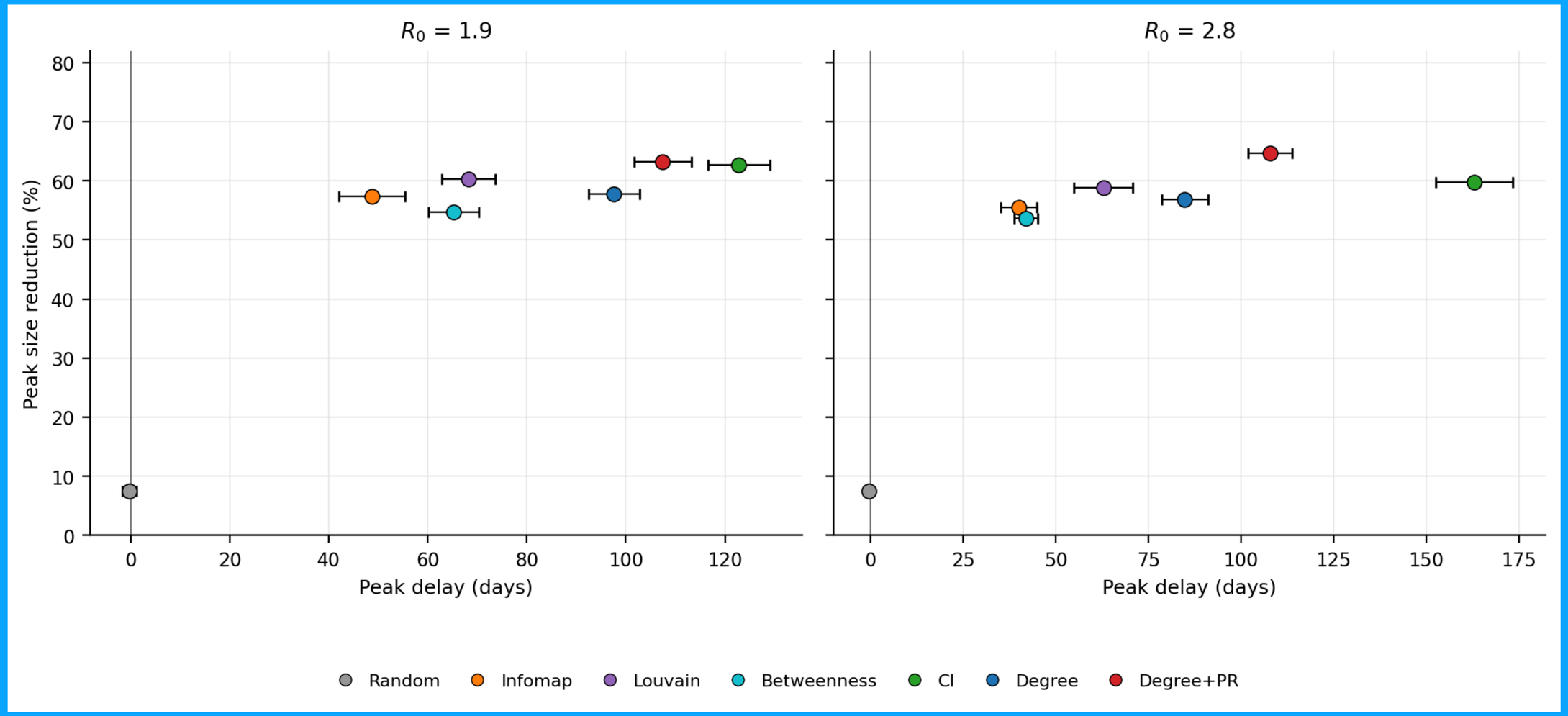
RESULTS – INFECTION REDUCTIONS

Measured on succesful outbreak runs only



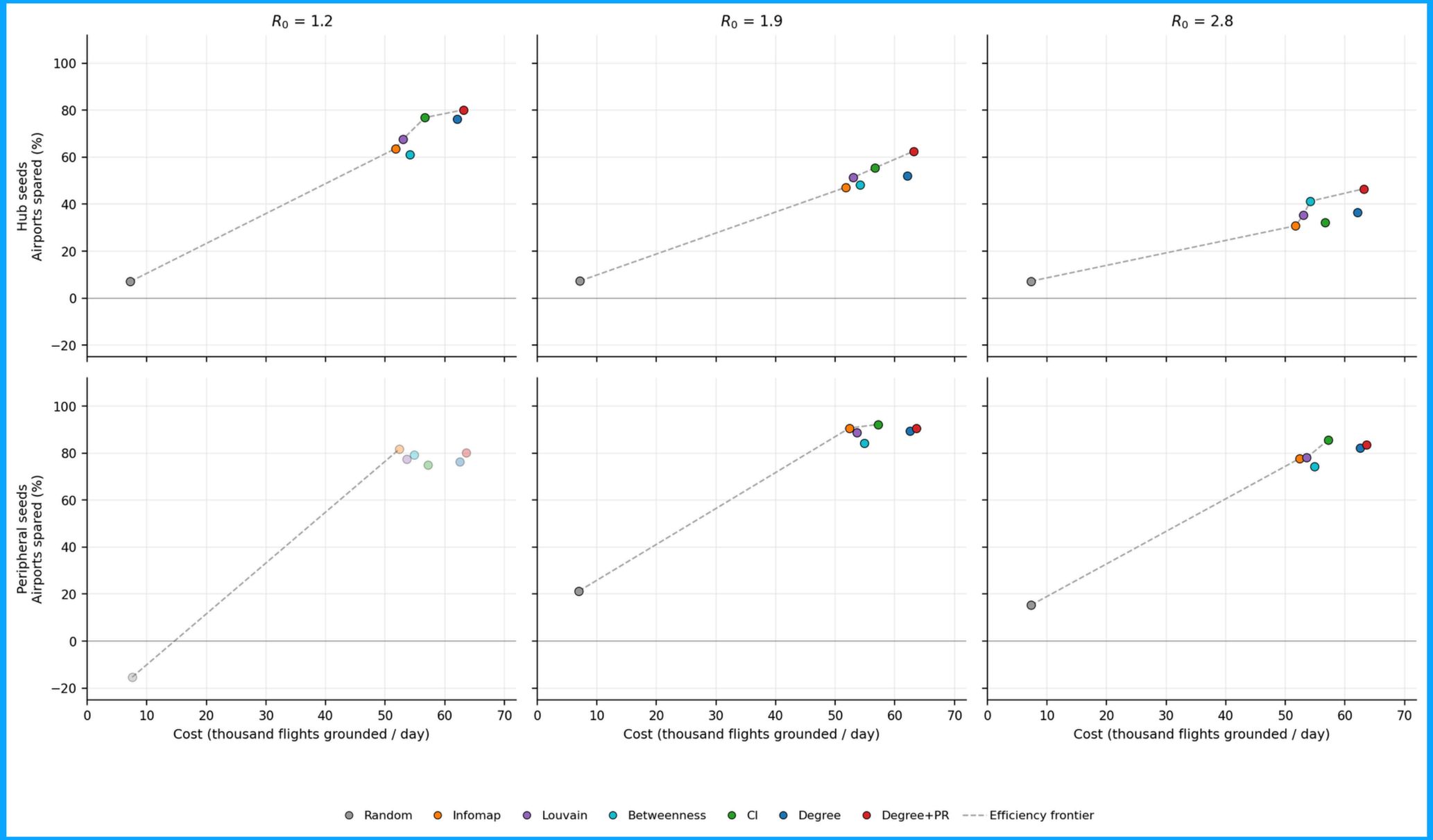
Degree+PR leads at every R_0. All strategies degrade as R_0 grows, and the gap between them narrows.

RESULTS – PEAK TIME INCREASE AND SIZE REDUCTION

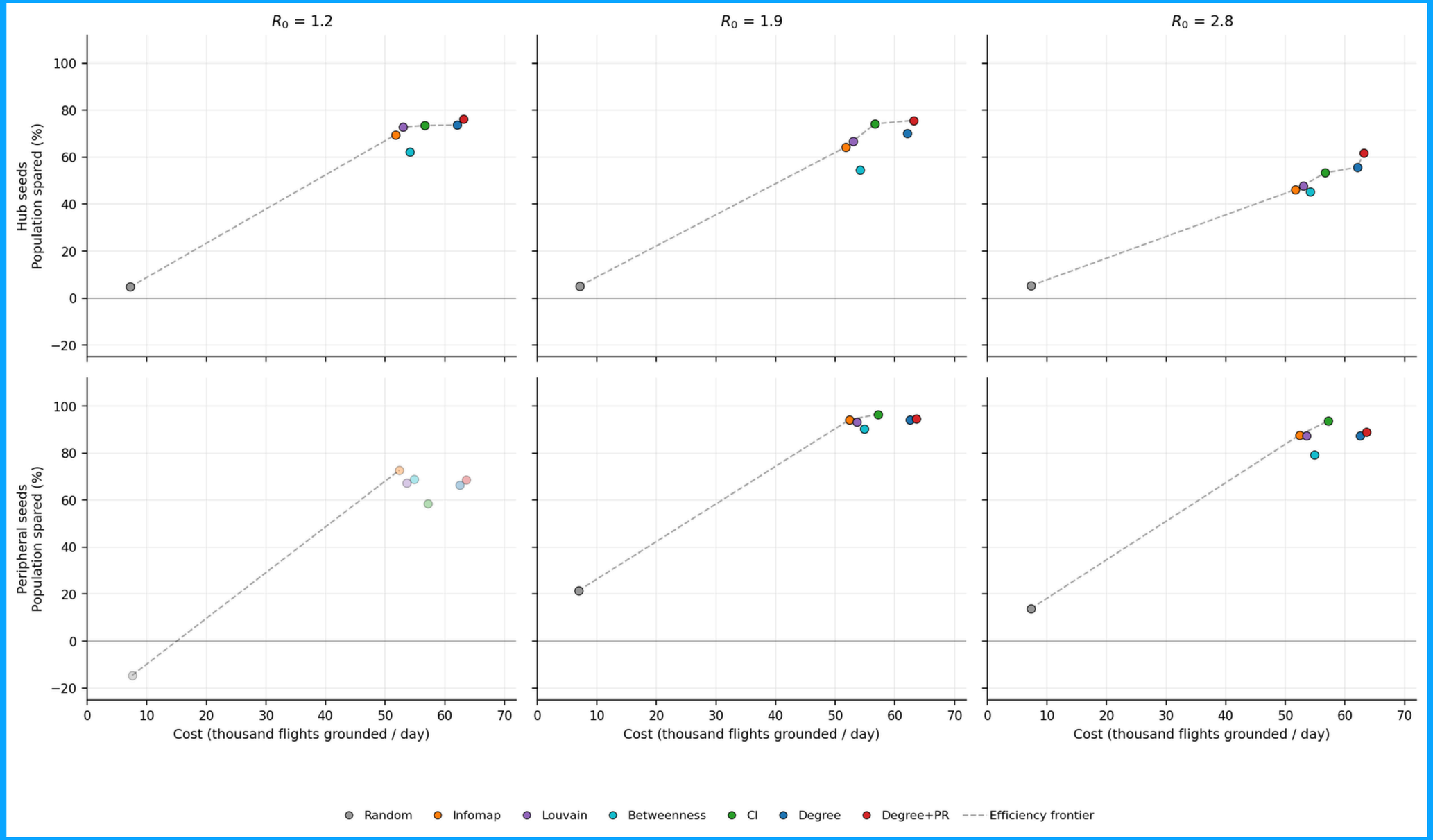


Hub seeds only, outbreak runs only and R0 of 1.2 removed since many baseline peaks were beyond 548 days. The **Collective Influence** approach wins over the weighted degree+PageRank strategy achieving **greater epidemic curve flattening**

RESULTS – MEASURES EFFICIENCIES FOR NODES



RESULTS – MEASURES EFFICIENCIES FOR POPULATION



ANSWERS TO RESEARCH QUESTIONS

What are the characteristics of the World Airport Network? Does it have any interesting properties?

- The World Airport Network presents a truncated power-law degree distribution and its positive assortativity can be read as a sum of two different “resolutions”

Is hub targeting the best way to obstruct a disease spread? Are there other preferable approaches?

- Targeting nodes with high weighted degree centrality is always suboptimal with respect to the strategy that takes PageRank into account. The CI algorithm is best with respect to peak delay and reduction.

How do our results change if we factor in the economic aspect of shutting down entire airports?

- It is possible to build an efficency frontier to understand which strategies are dominated. All the strategies on the frontier mean that if you spend more you will get more protection, but the steeper the line the more it makes sense to switch the strategy



CONCLUSION



There is no single best answer for all scenarios

Knowledge on disease aggressiveness and starting points is key to selecting the most effective strategy

Thank you for the kind attention!

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